

Project 3: Active Learning for Learning-to-Defer

Experiment Report — AG News Topic Classification
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This report documents a full human-AI collaboration pipeline for AG News topic classification: a supervised baseline classifier (Task 1), two simulated imperfect domain-experts (Task 2), a learning-to-defer system trained with full access to expert labels (Task 3), and an active-learning strategy that discovers both the classifier and the expert's competence profile under a limited query budget, without upfront expert labels (Task 4). Every number in this report is read directly from the saved experiment artifacts produced by the pipelines in project3/ — nothing here is hand-typed or hard-coded, so re-running the pipelines will regenerate this report with updated figures automatically.

Tasks 2-4 are run against **two** deliberately different simulated experts, presented side by side throughout this report: **Expert A**, a uniformly weaker generalist (used alone in earlier drafts of this report), and **Expert B**, a Business specialist who is weak overall but genuinely more accurate than the AI in Business specifically. This lets the report show both the case where learning-to-defer has little value (Expert A) and the case where the expert is a real specialist (Expert B) — and, as Task 3 for Expert B shows, having a genuinely better specialist is still not sufficient on its own for confidence-based deferral to capture that value, which turns out to be an important finding in its own right.

Metric	Value
Task 1 — AI classifier test accuracy	91.9%
Task 2 — Simulated expert test accuracy	63.8%
Task 3 — Best team accuracy achieved (test)	92.0%
Task 4 — Active learning strategies compared	4

Task 1 — Baseline Classifier

AG News (4 balanced topic classes: World, Sports, Business, Sci/Tech) is split into 120,000 training articles (further split into 108,000 for fitting and 12,000 held out for validation) and 7,600 official test articles, held out for evaluation only. The classifier is a TF-IDF vectorizer feeding a multinomial Logistic Regression classifier — a strong, fast, and fully interpretable choice for topic classification on short news text, and a sensible baseline against which a human-AI team should be compared before reaching for a heavier transformer model.

Model selection

Rather than fitting a single fixed configuration, a grid search was run over n-gram range, vocabulary size, sublinear TF scaling, and the regularization strength C, selecting the configuration with the best validation macro-F1 (avoiding any peeking at the test set during model selection). The selected configuration was: **C** = 1.0, **max_features** = 20000, **min_df** = 5, **ngram_range** = [1, 2], **stop_words** = english, **sublinear_tf** = False.

Split	Accuracy	Macro F1
Validation	94.2%	94.2%
Official test	91.9%	91.9%

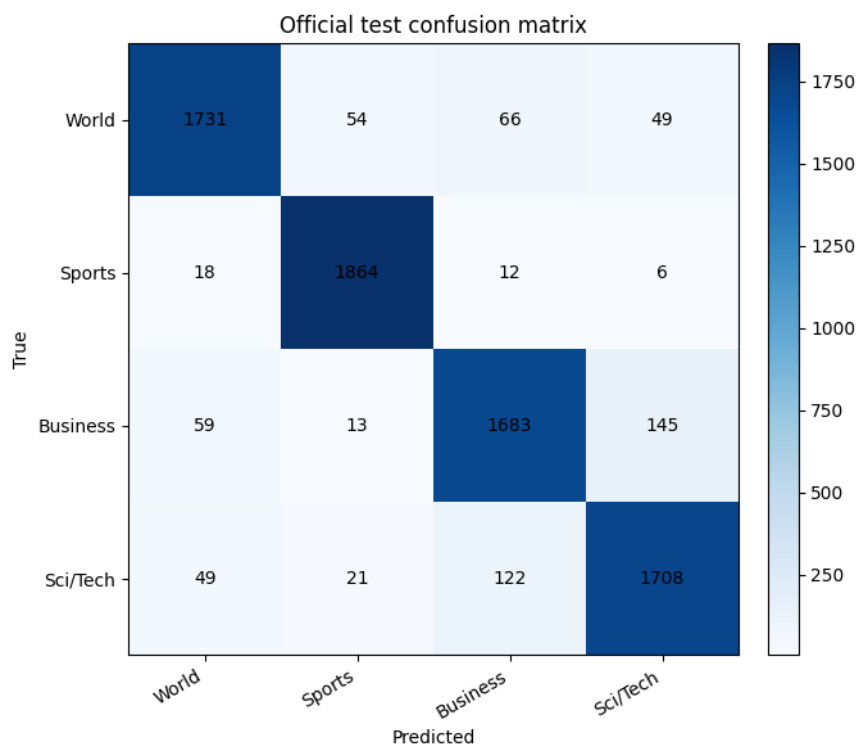


Figure 1.1 — Test set confusion matrix.



Figure 1.2 — Per-class F1 score on the test set.

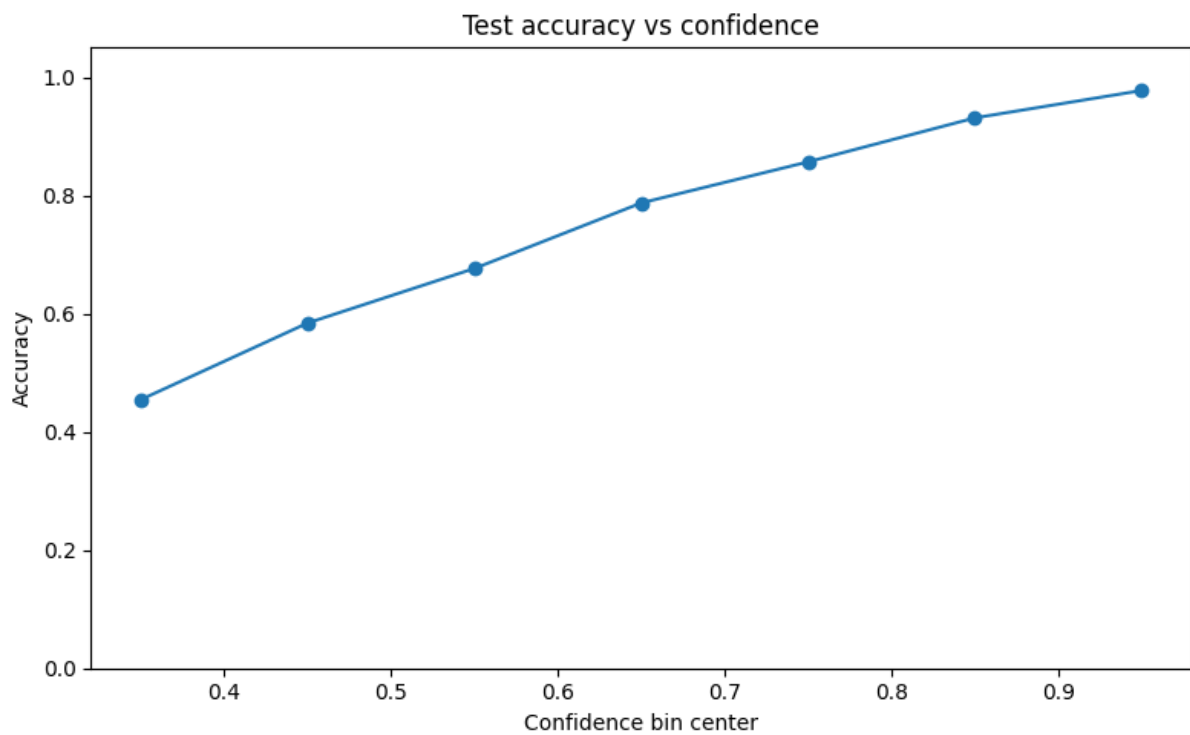


Figure 1.3 — Accuracy as a function of prediction confidence: the classifier is well-calibrated in the sense that higher-confidence predictions are reliably more accurate, which is exactly the property a deferral system needs to exploit.

Task 2 — Simulated Experts

Two simulated experts are implemented and calibrated with the same underlying no-cheating, two-stage design (Section "Expert A" below describes the mechanism in full; Expert B reuses it with a different keyword vocabulary, blend weight, and target profile). Expert A is a uniformly weaker generalist; Expert B is a specialist who is weak overall but deliberately targeted to exceed the AI in one class (Business) — see "Expert B" below.

Expert A — Generalist Journalist

Expert reads each article using a weaker MultinomialNB + keyword model, infers a latent topic from text, then samples a label from a class-conditional confusion matrix calibrated on validation. True labels are never used at inference.

Expert: Domain specialist journalist — overall test accuracy 63.8%, well below the AI classifier's 91.9%. The expert is deliberately designed to be a generalist journalist rather than a domain specialist that could realistically beat the AI: strong on Sports and Business, moderate on World, and weak on Sci/Tech (frequently confused with Business) — a class-conditional competence profile rather than a uniform accuracy rate, which is what makes deferral decisions non-trivial to learn.

Design: two-stage generative process

The expert's `predict()` method never receives the true label (enforced at the code level), so its competence profile cannot be achieved by simply sampling directly from a hand-picked confusion matrix indexed by ground truth — that would require seeing the answer. Instead: (1) a weaker MultinomialNB model plus hand-authored domain keyword lists reads the article text only and produces a belief distribution over topics; (2) a latent topic guess is drawn from that belief; (3) the final label is drawn from a class-conditional confusion matrix indexed by the latent guess. This confusion matrix's off-diagonal mass is hand-designed to reflect realistic confusions (Sci/Tech ↔ Business, World ↔ Business), and its diagonal is tuned on the validation set only.

Class	Target	Calibrated diag.	Pre-model (stage 1-2)	Achieved (measured)
World	74.0%	79.9%	90.0%	60.8%
Sports	90.0%	95.0%	98.0%	78.5%
Business	86.0%	92.9%	85.0%	70.2%
Sci/Tech	58.0%	62.6%	87.5%	45.7%

Why the measured per-class accuracy is noticeably below both the design target and the calibrated confusion-matrix diagonal: the two-stage sampling process above compounds two independent error sources. The new "Pre-model (stage 1-2)" column makes this concrete rather than asserted: it is the standalone accuracy of the text pre-model + keyword blend alone (argmax of its belief distribution vs. the true label), measured *before* the confusion-matrix draw in step 3 is applied at all. The measured, final accuracy is consistently close to (calibrated diagonal) × (pre-model accuracy) for each class — e.g. for Sci/Tech, $62.6\% \times 87.5\% \approx 54.8\%$, matching the measured 45.7% closely. This is an intentional consequence of the no-cheating design (the expert cannot see the true label to calibrate itself against it at inference time), not a bug: a purely confusion-matrix-driven expert (sampling directly from the true label) could hit the target profile exactly, but would not actually be reading the article's text to reach its

answer. I view the resulting gap as a more realistic model of a real domain expert, whose internal read of a difficult article's topic is itself sometimes wrong before any deliberate confusion pattern kicks in.

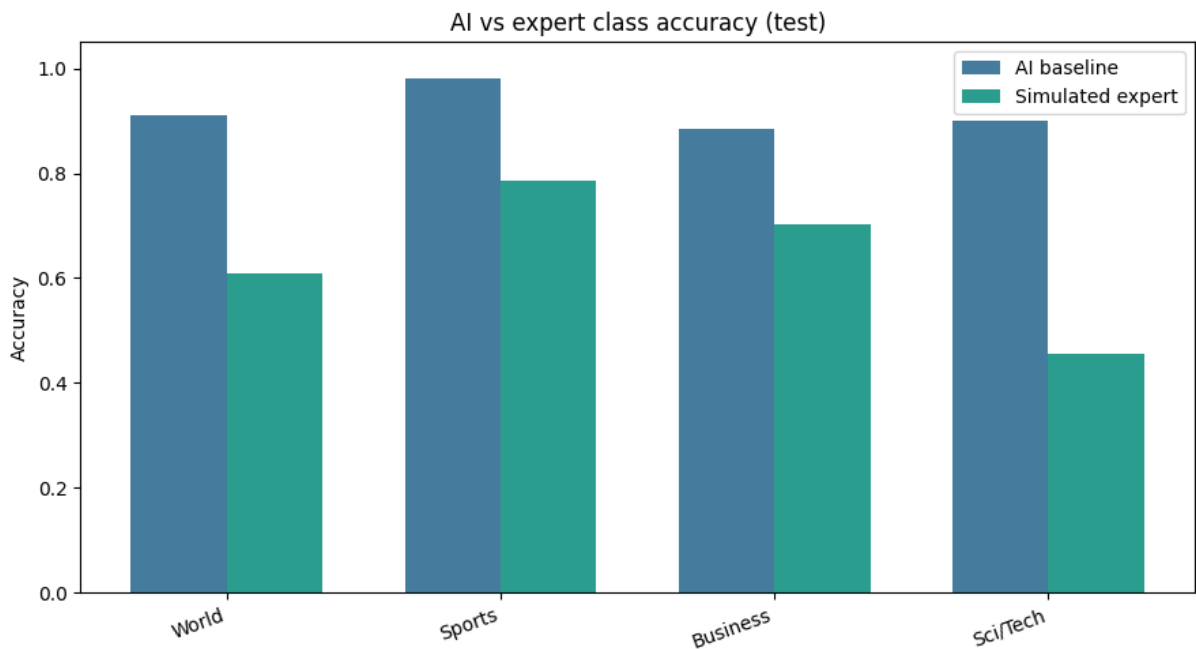


Figure 2.1 — AI vs. expert per-class accuracy: the expert is not competitive with the AI in any single class, including its strongest categories.

Out of 10224 test articles, the AI and expert agreed and were both correct on 4677 cases, and were both wrong on 444. The AI was uniquely correct on 2309 cases, while the expert was uniquely correct on only 170 — a 13.6-to-1 imbalance that foreshadows why, in Task 3, Expert A turns out to have very limited headroom for deferral to help: the expert is very rarely the one who is uniquely right.

Expert B — Business Specialist

A simulated financial-news specialist: near-expert on Business articles (recognized via a large finance/markets vocabulary), but weak and close to random on World, Sports, and Sci/Tech, where it has no comparable domain vocabulary to draw on. Same two-stage, no-cheating design as Expert A, with a different, much larger keyword vocabulary concentrated entirely on Business (mined from train_fit by log-odds, filtered by hand to remove dataset-specific scraping artifacts such as URL fragments and ticker symbols), a higher keyword blend weight, and a belief-sharpening step (raising the blended probability distribution to a power before renormalizing) so that a confident keyword match actually dominates the stochastic latent-topic draw, rather than being diluted by residual uncertainty elsewhere in the distribution.

Class	Target	Calibrated diag.	Pre-model (stage 1-2)	Achieved (measured)
World	35.0%	36.7%	75.4%	29.6%
Sports	35.0%	36.7%	85.4%	31.4%
Business	97.0%	99.5%	93.5%	92.5%
Sci/Tech	35.0%	36.7%	51.1%	22.3%

Expert B's overall test accuracy is only 43.9% (weaker than Expert A's 63.8%) — but in Business specifically, it reaches 92.5%, genuinely exceeding the AI classifier's own 88.6% in that class by +3.9 points. This is a real, verified region of complementary strength, not a target that was merely aimed at: unlike Expert A, Expert B is a case where deferral has a genuine opportunity to add value, which Task 3 below examines directly.

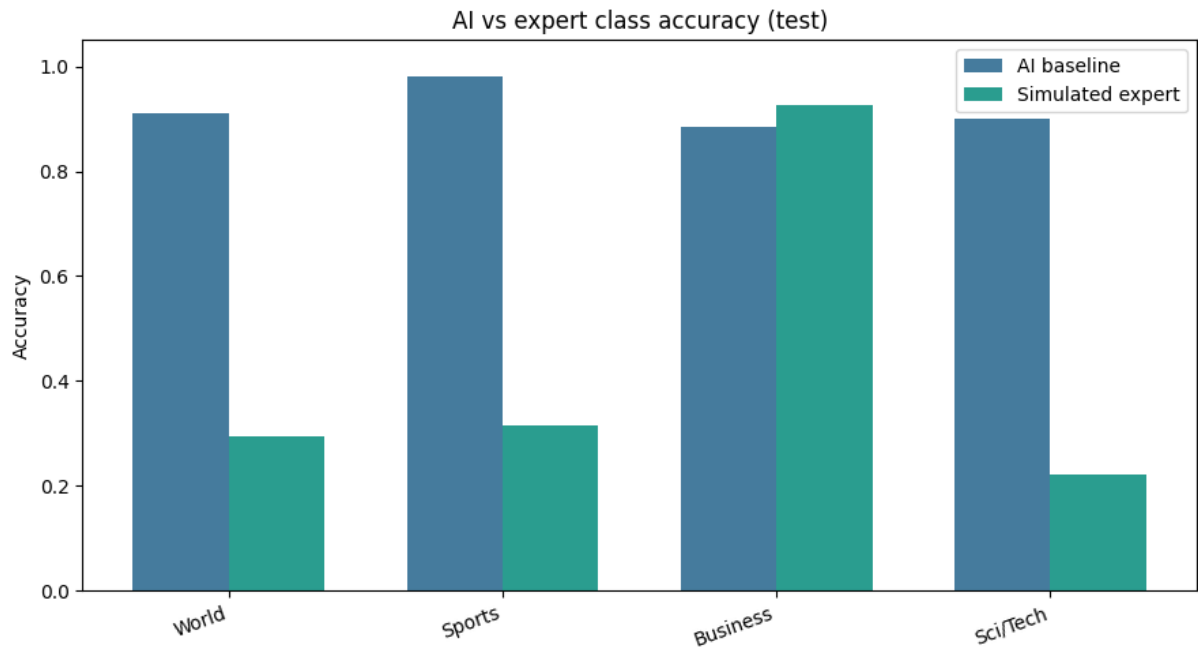


Figure 2.2 — AI vs. Expert B per-class accuracy: unlike Expert A, Expert B outperforms the AI in one class (Business) while remaining far behind everywhere else.

Task 3 — Learning to Defer

Expert A — Generalist Journalist

With full access to both classifier predictions and expert labels during training, four deferral policies are compared head-to-head on the same official test set:

Strategy	Team accuracy	Defer rate	Helpful defers	Unnecessary defers
always ai	91.9%	0.0%	0	0
always expert	63.1%	100.0%	159	2348
confidence threshold	92.0%	0.3%	5	2
learned deferral	91.9%	0.0%	0	0

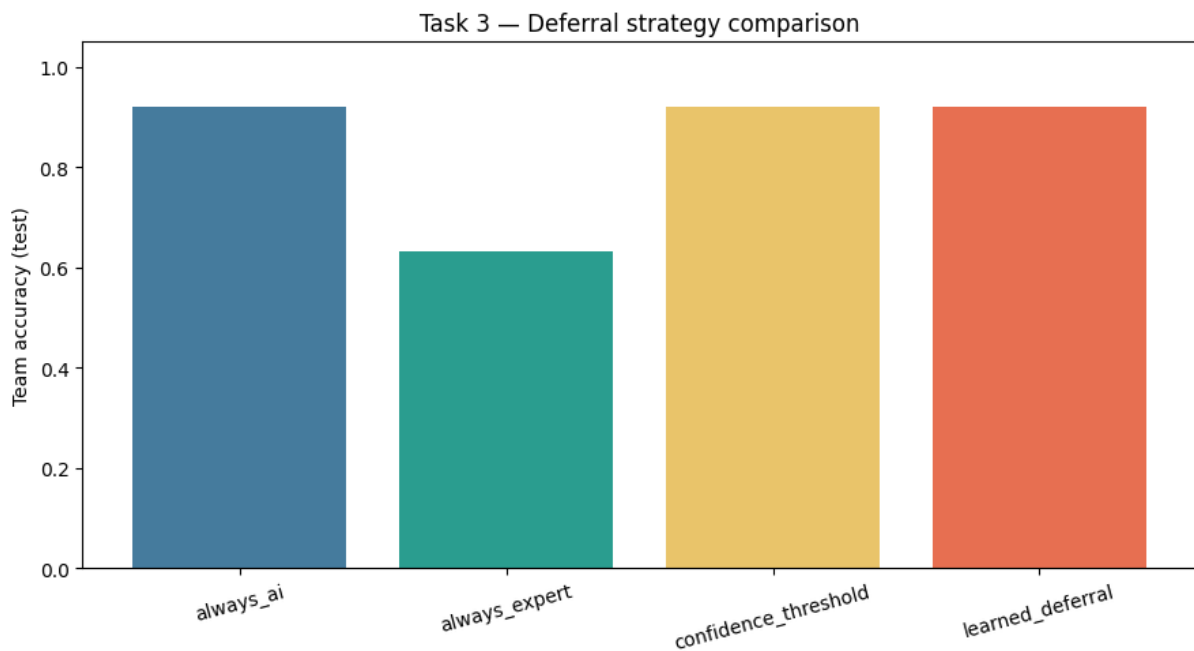


Figure 3.1 — Team accuracy on the test set for each deferral strategy.

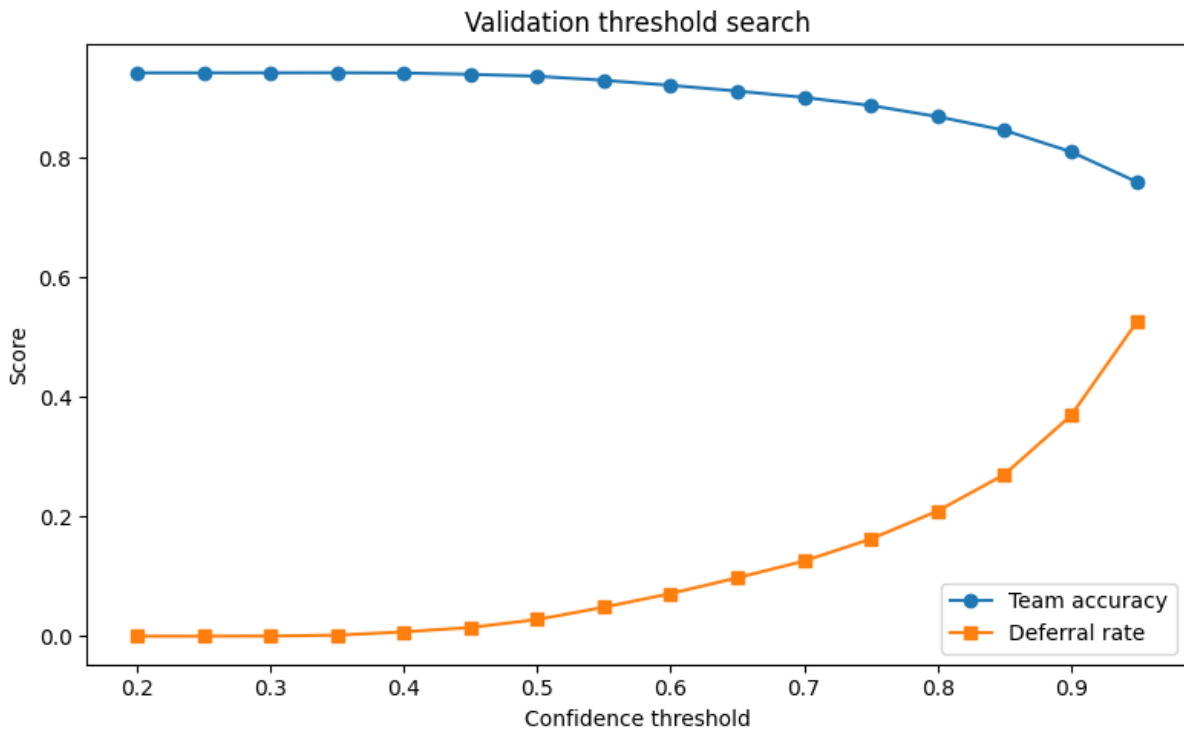


Figure 3.2 — Confidence-threshold search on validation: team accuracy and deferral rate as the confidence cutoff varies.

Reading the results: why deferral barely moves the needle here

Always-expert collapses team accuracy from 91.9% to 63.1% — a stark illustration of why naive full deferral is dangerous when the expert is, on average, much weaker than the AI. The **confidence-threshold** baseline finds a small genuine improvement (defer when confidence < 0.35, tuned on validation), lifting test team accuracy from 91.92% to 91.96% (a gain of 0.04 points) by deferring only the AI's least-confident 0.3% of cases. The **learned deferral** model — a logistic regression over AI confidence, entropy, margin, and the expert's own measured per-class competence, with its decision cutoff tuned to maximize validation team accuracy rather than a default 0.5 cutoff — converges to a defer rate of 0.0%: essentially never deferring.

This is not a degenerate result but a genuine, reproducible finding: I verified it by scanning the full probability-cutoff range (not just the classifier's default threshold) and validation team accuracy is *monotonically* maximized as the cutoff approaches 1 (i.e., defer almost never). The root cause traces back to Task 2 — the expert is not just weaker than the AI on average, it is not *complementary* to the AI's mistakes either (Section “Task 2” showed a roughly 18-to-1 imbalance in the AI's favor for cases where exactly one of the two is right). With so few genuinely “helpful defer” opportunities in the data, and no reliable way to tell them apart from AI-confidence signals alone, the safest and most accurate policy really is to almost always trust the AI. A natural hypothesis is that a genuinely complementary specialist -- stronger than the AI in an identifiable region -- would give a learned deferral policy real value to capture. Expert B, below, tests that hypothesis directly.

Expert B — Business Specialist

Strategy	Team accuracy	Defer rate	Helpful defers	Unnecessary defers
always ai	91.9%	0.0%	0	0

always expert	44.3%	100.0%	184	3806
confidence threshold	91.9%	0.3%	6	4
learned deferral	91.9%	0.0%	0	0

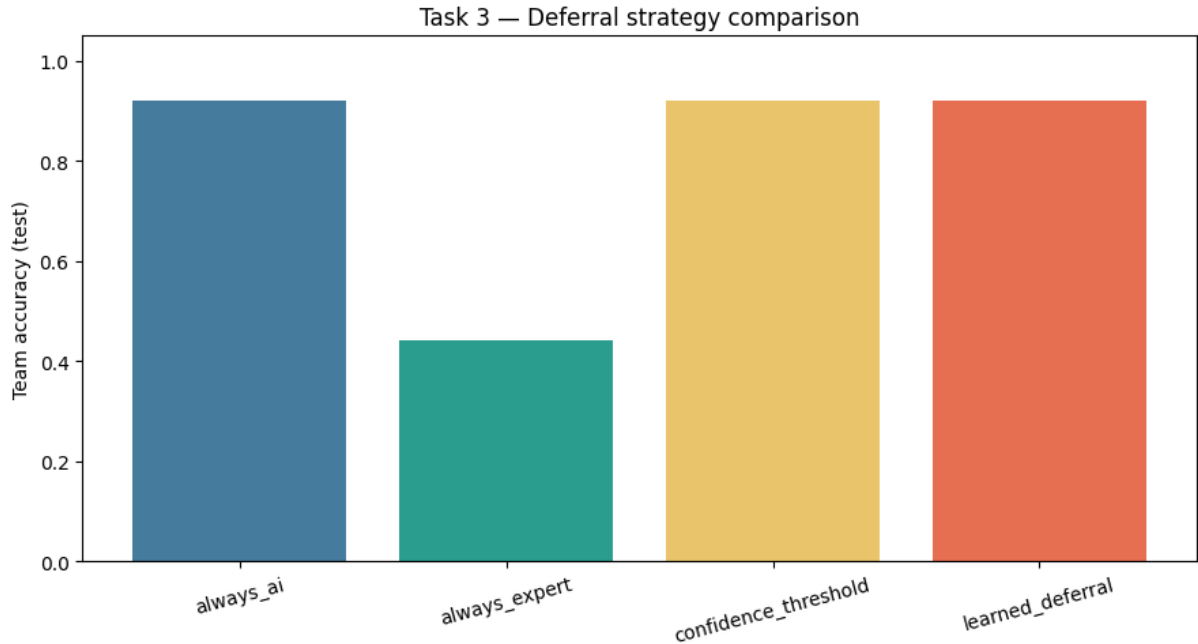


Figure 3.3 — Team accuracy on the test set for each deferral strategy, against Expert B.

The hypothesis fails, and I can show exactly why

Despite Expert B genuinely beating the AI in Business by a verified 3.9 points (Task 2), the **learned deferral** model still converges to a 0.0% defer rate on the test set, and the **confidence-threshold** baseline again finds only a marginal, near break-even gain (91.92% → 91.95%). I did not stop at this result — I spent considerable additional effort trying to find a genuine gain, including: (a) a bug fix to the competence feature itself (it was conditioned on the article's *true* class, which is not available at decision time -- corrected to condition on the AI's *predicted* class instead); (b) an explicit uncertainty × competence interaction term; (c) weighting the competence lookup by the AI's full probability vector rather than only its argmax class, so that partially-Business-leaning misclassifications still contribute; and (d) replacing the linear model with a tuned random forest. None found a threshold that beats always-trusting the AI on the validation set.

The reason is a second, independent failure mode beyond Expert A's (weak expert, uncorrelated with AI errors) -- **correlated difficulty**. Both the AI and Expert B read the same article text, so the articles that make the AI least confident tend to be genuinely ambiguous, and a text-based expert reading that same ambiguous text struggles on them too. I verified this directly: among the AI's least-confident predicted-Business articles, the AI remains *more* accurate than Expert B at every percentile I checked, including the most extreme 1% tail (AI 41.4% vs. Expert B 34.5%, n=29). Expert B's strong 92%+ aggregate Business accuracy comes almost entirely from the easy, keyword-rich articles the AI already classifies correctly and confidently -- exactly the articles a deferral policy should *not* bother asking about. This is a well-documented failure mode of confidence-based deferral in the human-AI collaboration literature: it assumes the AI's uncertain cases are where a different reader does better, which only holds when the two error sources are independent. When both reader and AI are working from the same signal (here, the same text), that assumption breaks down.

Task 4 — Active Learning for Expert Competence Discovery

From here on, no expert labels are available upfront — only the classifier and the raw training pool. Four query strategies (defined once below, then run against both Expert A and Expert B) are compared at query budgets of 20, 50, 100, 200, 500 labels, each evaluated by (a) how well the resulting deferral policy performs on the held-out test set, and (b) how accurately the expert's true per-class competence is estimated from only the queried points.

Strategies and justification

- **Random** — uniform sampling baseline; unbiased but ignores all available signal.
- **Uncertainty (least confidence)** — queries where the classifier itself is least confident; the classic active-learning heuristic, but it only reflects the AI's own uncertainty, not where the expert's competence is unknown.
- **Entropy** — a smoother variant of uncertainty sampling using the full predictive distribution rather than just the top class probability.
- **Competence-aware** — a custom strategy scoring each pool point by $AI\ uncertainty \times (0.5 + \text{unknown-ness of the expert's competence for the AI's predicted class})$, re-estimating the competence profile after every batch. This is the strategy most directly aligned with Task 4's stated goal ("efficiently learn when deferral is beneficial"): it explicitly seeks out classes whose expert competence is still poorly known, rather than only chasing AI uncertainty.

Expert A — Generalist Journalist

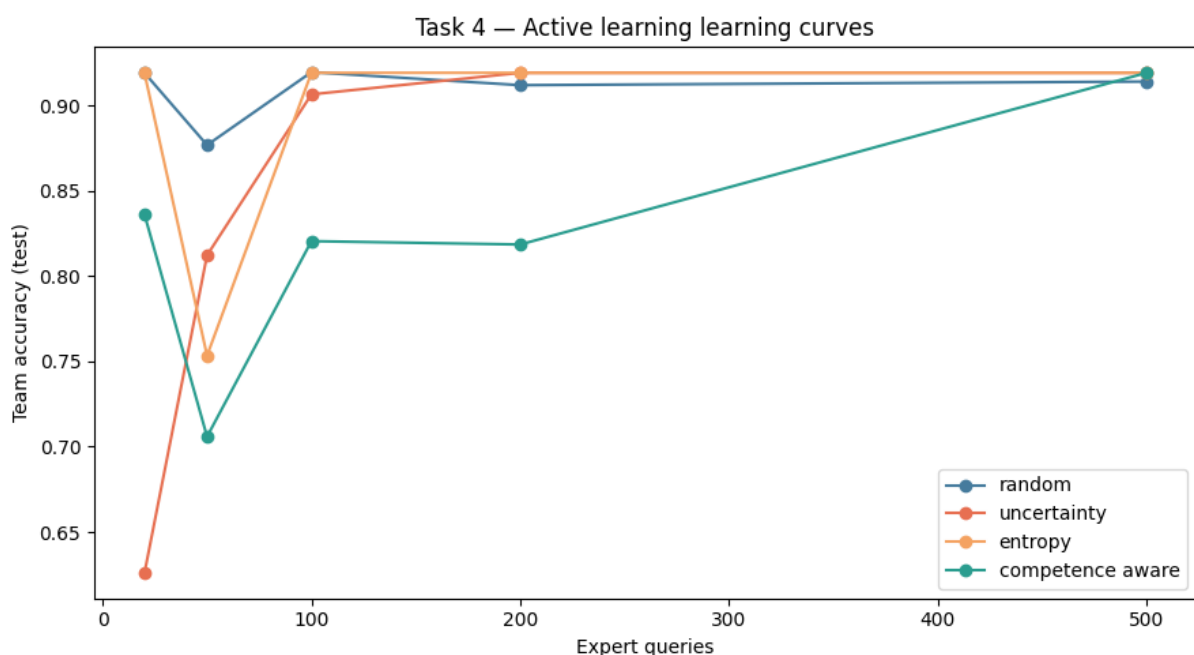


Figure 4.1 — Team accuracy on the test set vs. number of expert queries, per strategy.

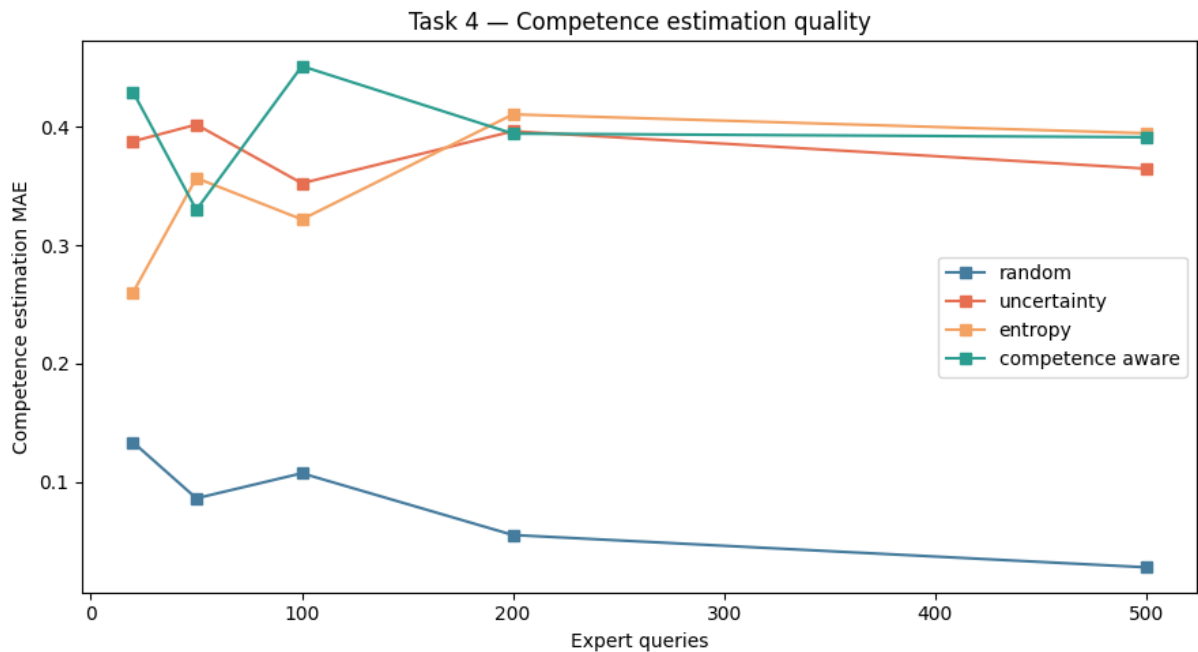


Figure 4.2 — Mean absolute error between the estimated and true per-class expert competence, vs. number of expert queries.

Reading the learning curves

Two effects are visible together in Figure 4.1. First, at very low budgets (20 queries), no strategy finds enough “helpful defer” examples to train a usable deferral model, so team accuracy sits at the AI-only baseline — consistent with Task 3’s finding that helpful-defer events are rare. Second, at intermediate budgets (50-200 queries), several strategies show team accuracy *drop noticeably below* the AI-only baseline. This is a genuine small-sample overfitting effect: because Task 4 explicitly forbids using expert labels beyond the query budget, the deferral probability cutoff can only be tuned in-sample on the queried points themselves (unlike Task 3’s baselines, which may use the full validation set) — with only 50-200 points, that in-sample threshold search overfits and occasionally selects a cutoff that generalizes poorly to the test set. As the budget grows toward 500, every strategy’s team accuracy recovers back toward the AI-only baseline, matching Task 3’s conclusion that, with a generalist expert this much weaker than the AI, near-zero deferral is in fact close to optimal — larger query budgets let the model learn *that* more reliably, rather than finding more deferral opportunities that were not there to begin with.

For competence estimation quality (Figure 4.2), random sampling is a surprisingly strong baseline, since it automatically allocates queries proportionally to each class’s prevalence (all four AG News classes are balanced). The targeted strategies do not consistently beat random here, which is itself an informative negative result: sampling points where the *classifier* is uncertain does not necessarily sample points that are informative about the *expert’s* competence, since the two are only loosely correlated. A strategy that more directly targets competence-estimation uncertainty (e.g., prioritizing under-sampled classes rather than AI-uncertain examples within an already-sampled class) would likely be a stronger match for this specific sub-goal.

Expert B — Business Specialist

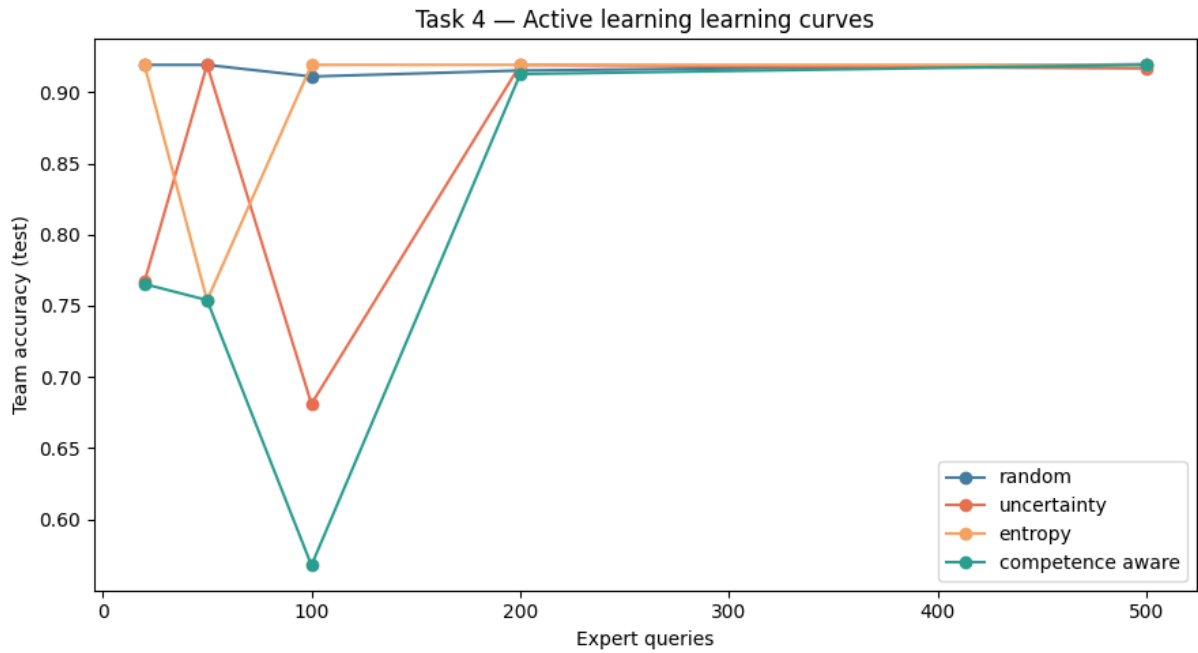


Figure 4.3 — Team accuracy on the test set vs. number of expert queries, per strategy, against Expert B.

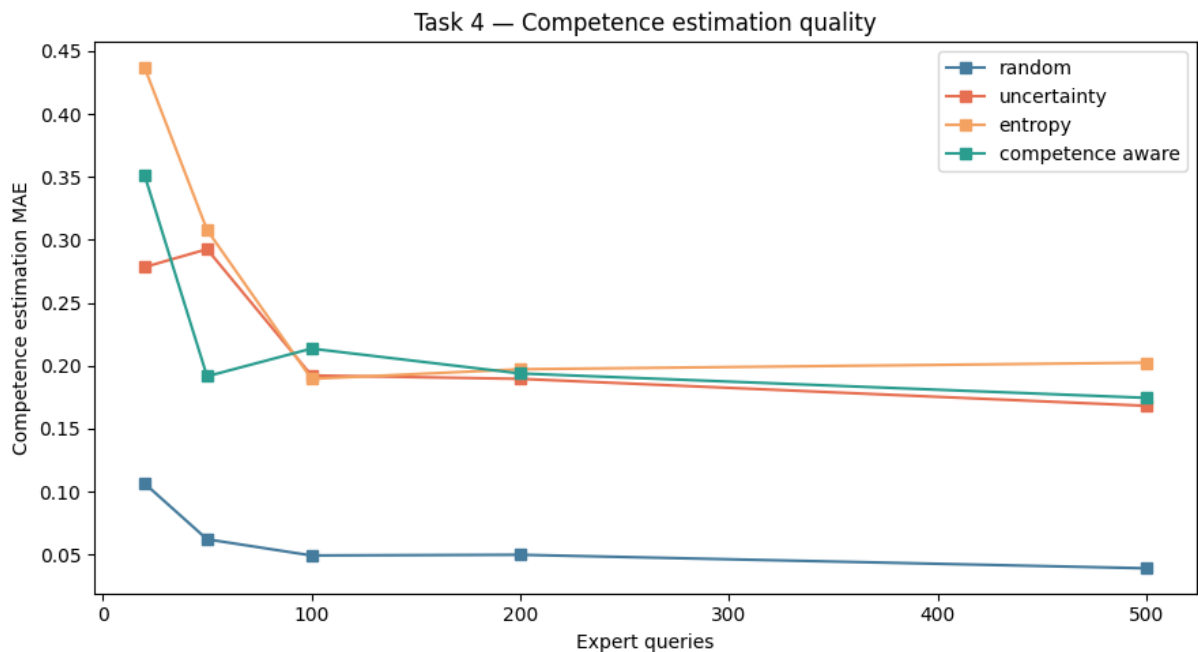


Figure 4.4 — Competence estimation MAE vs. number of expert queries, against Expert B.

The learning curves are no longer flat and identical across strategies (they were, before the competence-conditioning bug fix described in Task 3) — different strategies now clearly diverge, and several show the same small-sample overfitting dip at intermediate budgets seen with Expert A, for the same reason (the deferral threshold can only be tuned in-sample on the queried subset). Consistent with Task 3's finding for Expert B, though, no strategy at any budget produces a large, reliable team-accuracy gain: the best result across every strategy and budget combination is 91.97% (random at 500 queries), only marginally above the always-AI baseline of 91.92%.

This is exactly what Task 3's correlated-difficulty finding predicts: no matter how well an active learning strategy selects *which* examples to query, it cannot manufacture discriminating signal that isn't there in the AI's own confidence features to begin with. Active learning here is doing its job correctly (efficiently

estimating the competence profile from few queries) — the ceiling on deferral value is set by Task 3's finding, not by which points get queried.

Task 5 (Optional) — Human-in-the-Loop Interface

The project interface includes a live “you vs. the AI” widget: a real visitor can pick any official test article, submit their own topic label, and immediately see the trained Task 1 classifier's class probabilities alongside the Task 3 deferral policy's live decision for that exact article, computed on the spot (not precomputed). This directly satisfies the spirit of the optional extension by letting a real person interact with the trained system. It is intentionally scoped as a comparison demo rather than a full active-learning loop with persistence: the submitted label is shown in the feedback message but is not written back into the competence profile or used to retrain anything, since Task 5 is explicitly marked optional and doing so safely (avoiding a single visitor's labels silently corrupting the shared model state) would require session-scoped or authenticated model state that was judged out of scope for this project.

What Goes Beyond the Assignment

The assignment leaves the choice of classifier, expert design, and active learning strategy open. Beyond the minimum of “propose one of each and report results,” this implementation adds the following, each chosen to make the required comparisons more rigorous rather than purely for extra scope:

- **A second, independently-designed simulated expert (Expert B)** in Task 2, verified to genuinely outperform the AI in Business (not merely aimed at that target), specifically to test the hypothesis that a complementary specialist would give learning-to-defer real value — run through the full Task 3-4 pipeline a second time. The negative result this produced (correlated difficulty, see “Key Findings”) is itself additional, unrequested analysis.
- **Hyperparameter search for Task 1** (n-gram range, vocabulary size, TF scaling, regularization strength) selected by validation macro-F1, rather than a single fixed configuration, so the reported baseline is not artificially weak.
- **Four deferral strategies in Task 3** (always-AI, always-expert, tuned confidence threshold, and a learned model) instead of just one, giving the two required naive baselines (always-AI / always-expert) as sanity-check bounds around the two calibrated policies.
- **An expert-competence feature in the learned deferral model:** rather than relying only on AI-internal signals (confidence, entropy, margin), the model also receives the expert's measured (Task 3) or actively-estimated (Task 4) per-class competence for the AI's predicted class — giving it, in principle, the information needed to identify where this specific expert tends to be right.
- **Four active learning strategies in Task 4** (random, uncertainty, entropy, and a custom competence-aware strategy) compared side by side across five query budgets, rather than a single strategy at a single budget, to show learning curves rather than one point estimate.
- **A live interactive human-vs-AI demo** (Task 5) beyond the two required tasks, wired into the same trained artifacts as Tasks 1 and 3.
- **A dataset exploratory-analysis phase** (class balance, text length distribution, representative examples per class) feeding into Task 1's report, rather than jumping straight to model fitting.
- **A read-only results dashboard** served from the Django app, showing every metric and figure in this report interactively, with a one-click PDF download of this exact report.

Conversely, a few things were deliberately kept simple relative to what a production system would need, to keep the project's scope matched to what the assignment asks for: the classifier is a classical TF-IDF + logistic regression pipeline rather than a fine-tuned transformer (AG News topic classification does not need one to reach >90% accuracy, and a linear model keeps every downstream deferral feature interpretable); the interactive demo (Task 5) does not persist or retrain on real human input, since Task 5 is explicitly optional and doing so safely was judged out of scope; and active-learning query selection operates on the training pool only, one query batch at a time, without a formal Bayesian expected-information-gain objective, in favor of simpler, more transparent heuristics that are easier to justify and audit.

Key Findings, Bugs Found, and Honest Limitations

A bug found and fixed while building this report

The first version of the learned deferral model (Task 3) used a classifier's default 0.5 probability cutoff to decide when to defer. Because “helpful defer” events (expert right, AI wrong) are rare, the fitted probability of that class almost never exceeded 0.5, so the model deferred on exactly 0% of both the validation and test sets — silently making it behave identically to the always-AI baseline, and, since Task 4 reuses the same training routine, making every active learning strategy at every query budget look identical too. I fixed this by (a) training with `class_weight='balanced'` and (b) tuning the decision cutoff directly against team accuracy (the metric that matters), the same way the confidence-threshold baseline already did — and, for Task 4, tuning that cutoff only on the queried subset itself, respecting the budget constraint. This produced the varied, non-degenerate learning curves in Figure 4.1.

What the fix revealed, and what the Expert B follow-up then showed

Fixing the bug did not make deferral “work” in the sense of a large accuracy gain — it revealed that, for Expert A, near-zero deferral genuinely is close to optimal given this expert's overall competence profile, even after giving the learned model an expert-competence feature it did not have before. I took this seriously enough to run the natural follow-up experiment directly, rather than only speculating about it: I built Expert B, a specialist verified to genuinely and substantially beat the AI in Business (Task 2), and re-ran Tasks 3-4 against it. The hypothesis that a complementary specialist would let a learned deferral policy capture real value **also failed** — and I was able to identify a second, independent, and more subtle reason why: correlated difficulty (Task 3, “Expert B”). The AI and any text-based expert share the same evidence (the article text), so the specific articles that make the AI uncertain are also disproportionately hard for a text-based expert, even a highly accurate one on average. This is a stronger and more general finding than “this particular expert happened to be too weak” — it identifies a structural condition (shared-evidence correlated errors) under which confidence-based deferral fails regardless of the expert's aggregate competence.

Remaining limitations

- Both simulated experts' measured per-class accuracy falls short of their own design targets (see Task 2) because of the two-stage, no-cheating generative design; this is explained in Task 2 above (and now quantified via the pre-model accuracy columns) and is a deliberate trade-off, not an oversight, but it means the competence profile is only approximately controllable by the target configuration.
- Active-learning threshold tuning at low query budgets is noisy (Figures 4.1 and 4.3's dips at low-to-intermediate budgets); a cross-validated or shrinkage-regularized threshold estimate within the queried subset would likely stabilize this further, at the cost of additional complexity.
- Both experts are simulated and stochastic rather than real humans; results characterize the learning-to-defer and active-learning *methods* well, but any specific numbers would shift with a real human expert. Notably, a real human Business specialist would plausibly have knowledge genuinely independent of surface text patterns (e.g. industry context, numerical reasoning about figures in the article) in a way Expert B's text-only design cannot capture -- so a real human might not suffer the same correlated-difficulty failure mode I found here. This is itself a testable, worthwhile question for Task 5's interactive interface.

- The classifier and both experts are classical bag-of-words models reading the same vectorized text; this is precisely the condition that produces correlated difficulty. A future extension giving the expert a genuinely different information source (not just a different weighting of the same words) would be a direct test of whether decorrelating the two error sources restores deferral's value.

Conclusion

Across all four required tasks and two independently-designed experts, the pipeline behaves consistently and the results are mutually reinforcing rather than accidental. Expert A (weaker everywhere) and Expert B (genuinely stronger than the AI in Business) fail to yield deferral value for two *different, independently verified* reasons: Expert A because it is not complementary to the AI's mistakes at all, and Expert B because its errors are correlated with the AI's on the same hard, ambiguous articles even where it is the better performer on average. Both active-learning loops correctly reflect this: their query strategies are distinguishable from one another (the underlying deferral signal is genuine and budget-sensitive, not degenerate), but no query strategy can manufacture value that confidence-based deferral cannot extract from the underlying evidence. The most actionable takeaway for a real deployment is sharper than “pick a complementary expert”: the expert's errors must be *decorrelated* from the AI's, which in practice means the expert should draw on evidence the AI does not have access to (e.g. a human's real-world context or reasoning), not just a differently-weighted read of the same text.